School of Engineering and Material Science

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Control Systems Analysis and Design

Water Tank Level Control:
Analysis Using Different Control Systems

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Abstract

This paper focuses on the investigation of the effects of open and close loop control systems on water level control within a laboratory environment. This is carried out through an initial procedure, determining the time constant with an open loop system in place, and discussing the effectiveness of a simulation model in predicting the time constant obtained experimentally and via calculations, from which the simulation and experimental graphs were found to display a percentage error of 61.7% and 64.6% respectively. Following this, the procedure was changed to utilise a close loop, with a PID controller included. Three sets of PID values were utilised, with the aim to improve the output with each successive attempt, leading to a final tested set of values being P = 50, I = 0.5, and D = 3. These values led to the most optimised display of values, since the overshoot was minimised to 0.844% and 0.138% for the experimental and simulated plots respectively. Furthermore, the rise time also displayed a significant decrease when these PID values were applied. However, a small steady-state error arose, leading to the values still being improper for use in sensitive environments. Hence why, further optimisation is proposed through the use of the Ziegler-Nichols method to find the optimum PID values.

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1 Introduction

1.1 Experiment aims and objectives

Water level control is a crucial component in the chemical, food and beverages, and wastewater treatment industries. They are implemented to provide various benefits to these industries, which include an improved safety standard, as overflows and dry runs can be prevented, an increased efficiency resulting from increased optimisation of fluid use, and additional process stability, due to the control systems maintaining consistent fluid levels and ensuring reliable operation throughout the service lifetime [1]. However, PIDs have limitations due to the simplicity of their nature, reducing their effectiveness in complex systems. A critical aspect of this study is to investigate the effects of open and closed loop control systems on water level control within a tank. Open loop control describes a simple system in which the control action is independent of the desired output, as a result this increases its repeatability. Due to these reasons, they are most often utilised in straightforward applications such as automatic tea/coffee machines and electric hand dryers [2].

Close loop control is a modified version of the open loop control, within which there exists a feedback loop. Due to this, the feedback of the output is compared to a desired reference point, allowing for an error correction signal to be generated if any deviation from the reference exists. These systems are more complicated and as a result they have slower response times, hence they are often used in systems such as cruise control and air conditioning units. PID controllers often form part of a closed loop system, where they are used to adjust the output of a system based on the deviation between the setpoint of the system and actual variables [2].

The study consists of an initial simulation of both systems on MATLAB SIMULINK, with the aim to validate these findings through an experimental procedure, utilising the CE117 process trainer hardware and C2000 software to set up the open or close system, simulating real-world conditions and to record any signals. Through this, an insight into PID and control system design can be acquired, with further testing done to tune the PID in a closed system to optimise for the best system characteristics. The experimental data is then plotted and compared to the SIMULINK simulations determined previously, with a detailed analysis carried out to determine the differences. The following section details the system components and procedures carried out.

1.2 PID controller theory

1.2.1 Proportional response (K_p)

The proportional component (Figure 1) of the PID only depends on the difference between the setpoint voltage and the level transmitter (LT), which is the error term. Hence the proportional gain term is used to determine the ratio of the output response to the error signal generated. In general, increasing the proportional gain term would increase the speed of the response of the control system, however making this term too large can increase the oscillations of the system, making it unstable [3].

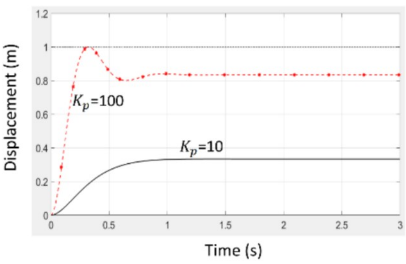

Figure 1: Effects of proportional response on a control system [4]

1.2.2 Integral response (K_i)

The integral component (Figure 2) of the PID functions to sum the error term over time, resulting in even a small error term causing the integral component to increase, continually increasing over time unless the error term is zero. Therefore the main effect of this term is to drive the steady-state error down to zero [3].

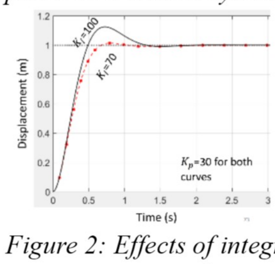

Figure 2: Effects of integral response on a control system [4]

1.2.3 Derivative response (K_d)

The derivative component (Figure 3) of the PID has the effect of causing the output to decrease if the process variable increases rapidly, since the response is proportional to the rate of change of the process variable. Hence why, increasing the derivative component has the effect of causing the system to react more strongly to changes in the error term, improving the speed of the overall system response. Most systems use a small derivative response term, since this term is very sensitive to noise, and can cause the system to grow unstable if the feedback loop is too noisy or slow [3].

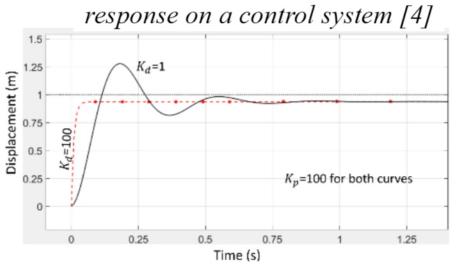

Figure 3: Effects of derivative response on a control system [4]

such as pressure changes within the system. These can significantly impact the performance of the system, which is further emphasised by the K_p value greatly increasing the flow rate, making frictional and pressure losses more prominent in the system. In addition to this, the overshoot was greatly impacted by the tuned PID values, leading to a minimal overshoot of just 0.844% in the experimental data, and 0.138% in the simulated data. The only downside to the large values of K_p and K_d is the introduction of a steady-state error of 0.02V, which while small can introduce potential risks in precision systems that cannot allow any steady-state errors. Due to these factors, it can be stated that the PID values could be further tuned and optimised, for this reason, the Ziegler-Nichols tuning methods can be suggested. These methods allow for a more direct approach in determining the PID values, hence a better tuned performance of the system.

5 Conclusion

In conclusion, the water level control system was analysed using both an open and close loop system to observe the response to a step voltage applied to the system, and measure the behaviour as the system settled. From the analyses conducted, it was found that the open loop system, both experimentally and simulated exhibited vast deviations to the calculated time constant of 111.33 secs. This shows how the calculation utilises many assumptions and cannot be used to accurately estimate the graphical time constant. This provides limitations in the use of certain methods to model the behaviour of the control system in a simulated setting, or estimate certain feature of a response. The close loop model provides a more suitable approach to level control, as the feedback loop provides a path for error correction, hence allowing the system to stabilise quicker and at a more suitable level, reducing the steady-state error of the system, which can be seen when comparing the open loop and close loop systems. The effects of PID values was also discussed in detail, with the results showing how the tuning of the PID values, more specifically increasing the K_p and K_d values, provided a significantly better system output in the experimentally obtained data, minimising the rise time and overshoot of the system, optimising to values of 12.2 secs and 0.844% respectively. This was further verified through the use of SIMULINK to find the same metrics, showing the correlation between the experimental and simulated data, showing the viability of using the simulations to accurately predict system performance.

6 References

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